TECHNICAL BULLETIN

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EPOXY CURING AGENT 106K
(ECA 106K)

Typical Properties

Property	Value	Test Method
Molecular Weight (average)	168-172	
Specific gravity @ 25°C, g/cc	1.205 ± 0.005	D-102 (ASTM D 4052)
Appearance	Yellow to amber liquid	D-104 (Visual)
Viscosity (Brookfield at 25 °C), cP	70-200	D-150F
Color, Pt-Co	300 - 700	D-103 (ASTM D 1209)
Color, Gardner	8 max	D-150G
Gel Time, min	34-47	D-160J

Applications

ECA 106K is a low viscosity liquid anhydride epoxy curing agent based on methyltetrahydrophthalic anhydride. For convenience, it is pre-catalyzed with an accelerator that contributes to good glass transition temperature (Tg) in cured polymers. It is designed for use in converting epoxy resins to highly crosslinked polymers with excellent physical and electrical properties. ECA 106K is used in demanding applications including laminating and filament winding. It can also be used in electrical applications such as casting, potting and encapsulation.

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ECA 106K
ECA 106K imparts ease of handling due to its low viscosity, low volatility and low freezing point. It is a liquid at room temperature. If crystallization does occur during shipping or storage, it can be returned to liquid by heating at 38-55°C (100-130°F) for 2-8 hours. Formulations of ECA 106K and liquid epoxies have a long pot life/working time in excess of 24 hours.

For more information on the use of anhydrides like ECA 106K as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical.

Formulation Example

ECA 106K was formulated with a variety of epoxy resins. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 106K, phr	Viscosity at 25 °C, cP	Gel Time [1], min	Tg [2], °C
Standard BPA Liquid Epoxy	89	850	41	122
Low Viscosity BPA Liquid	92	830	38	127
Cycloaliphatic Epoxy	122	250	39	189
Epoxy Phenol Novolac	96	1110	37	124
Epoxy BPA Novolac	87	3530	37	133
BPF Liquid Epoxy	96	435	32.6	129

1. Shyodu Hot Pot (100g at 100°C)
2. Cured 1 hr at 120°C and 1 hr at 220°C. Analyzed in DSC: 40°C/min to 250°C

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ECA 106K

Handling & Storage

ECA 106K will react with water to form diacids. This is normally undesirable, so ECA 106K should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 106K, consult the Dixie Chemical brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 106K is available from Dixie Chemical Company, Inc.'s plant in Pasadena, Texas. Contact your Dixie representative for details.